HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented automation workflows for equipment and tooling maintenance, reducing unscheduled downtime by **18%** across two production lines.
- Developed and executed validation strategies and comprehensive test plans for new automation systems, ensuring adherence to operational readiness standards.
- Led root-cause investigations for equipment failures, implementing corrective actions such as enhanced monitoring and predictive maintenance routines, which improved mean time between failures by **22%**.
- Managed configuration control for automation equipment and tooling, ensuring documentation accuracy and compliance with standard operating procedures.
- Mentored junior engineers on standardized best practices for automation system integration and troubleshooting, enhancing team efficiency in problem resolution.
- Coordinated cross-functional implementation efforts for automation deployments, facilitating readiness across manufacturing, quality, and operations teams.
- Contributed to continuous improvement initiatives by analyzing performance data and recommending data-driven optimizations for automated processes, increasing throughput by **12%**.

SKILLS

Automation & Controls – Robotics, Control Systems, Automation Principles, Equipment Reliability, Preventative Maintenance, Reliability Engineering

Programming & Troubleshooting – PLC, HMI, Python, SQL

Networking – TCP/IP, Routing, Switching, Network IDS/IPS, Active Directory, Firewalls

Data Analysis & Visualization – Statistical Tools, Data Visualization, Microsoft Excel, Microsoft PowerPoint

Engineering Practices – Change Control, Validation, Non-conformance, Corrective and Preventative Actions (CAPA), Technical Writing, Documentation Accuracy, Agile Methodologies

PROJECTS

Manufacturing Process Automation Enhancement (HCL Technologies)

- Automated a critical post-processing step in a manufacturing line using custom Python scripting, reducing manual intervention time by approximately **4 hours per shift**.
- Integrated vision system feedback into the automation control logic to improve defect detection accuracy by **15%**, leading to higher product quality.
- Developed comprehensive documentation and training materials for the automated process, enabling faster onboarding for new operational staff.

Industrial Control System Reliability Upgrade (HCL Technologies)

- Analyzed historical operational data from industrial control systems using SQL and Python to identify key drivers of equipment failure.
- Implemented enhanced diagnostic logging and alert mechanisms within the control system, which reduced critical alert response times by **30%**.
- Designed and tested interim solutions for common equipment faults, minimizing production disruption during repair periods.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)